Lecture 17: Barrier Methods

MA3236 Non-Linear Programming

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S17-08-21

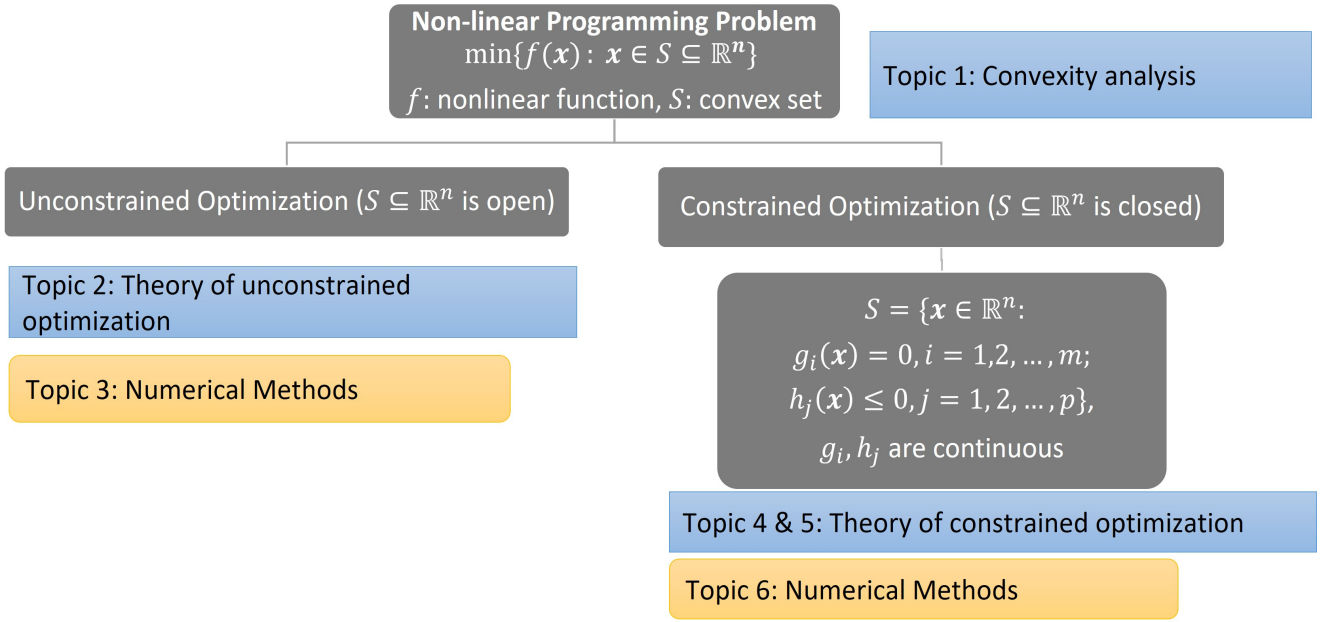
Quick review: our methods so far

Method	Inner subproblem	How constraints enter
QP	Unconstrained minimization	Quadratic penalty
ALM	Unconstrained minimization	Multiplier reference + penalty correction
SQP	Quadratic program	Linearized feasibility + Lagrangian curvature

- Studied mainly for equalities; extend to general constraints.
- Primal infeasible: iterates need not satisfy the original constraints.

Can we retain the feasibility for inequality constraints?

Topics



Topic 6: numerical methods for constrained optimization.

- Quadratic penalty and augmented Lagrangian methods
- Lagrange–Newton and sequential quadratic programming
- Barrier methods
- Duality, followed by algorithms for the dual problem

Constrained optimization with strict inequality feasibility

$$\min_{\mathbf{x}\in\mathcal{D}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{h}(\mathbf{x}) \leq \mathbf{0}.$$

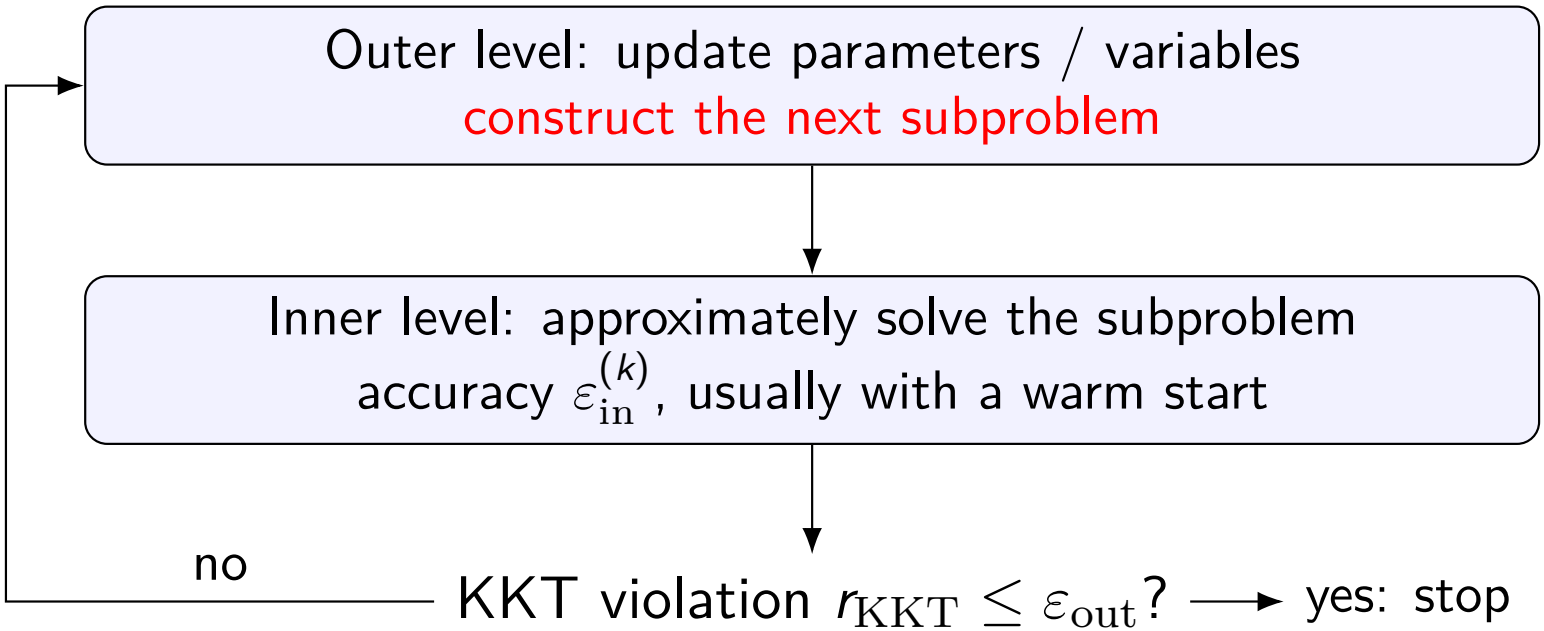
$$\mathcal{S}^\circ = \{\mathbf{x} \in \mathcal{D} : \mathbf{g}(\mathbf{x}) = \mathbf{0},\ \mathbf{h}(\mathbf{x}) < \mathbf{0}\} \neq \emptyset.$$

$\mathcal{S}^\circ$: strict inequalities on the equality set; need not be an ambient interior. Assume $\mathcal{D}$ open, C^2 data, and $p \geq 1$.

KKT conditions: with $\mathbf{A} = [\nabla g_1, \ldots, \nabla g_m]$ and $\mathbf{C} = [\nabla h_1, \ldots, \nabla h_p]$,

$$\begin{aligned} &\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu} = \mathbf{0}, \\ &\mathbf{g} = \mathbf{0}, \qquad \mathbf{h} \leq \mathbf{0}, \qquad \boldsymbol{\nu} \geq \mathbf{0}, \\ &\qquad\qquad \nu_j h_j = 0 \quad (j = 1, \ldots, p). \end{aligned}$$

Two-level iterative framework



This lecture: **barrier methods** **with** **constrained subproblems** **on** $\mathcal{S}^\circ$:

after initialization: keep $\mathbf{h} < \mathbf{0}$.

Constructing a wall at the boundary

Notice that

$$-h_j(\mathbf{x}) \downarrow 0 \quad \Longrightarrow \quad -\log(-h_j(\mathbf{x})) \longrightarrow +\infty.$$

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Add this boundary repulsion (barrier) to the objective:

$$\boxed{\Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu \sum_{j=1}^{p} \log(-h_j(\mathbf{x})), \qquad \mu > 0.}$$

$$\min_{\mathbf{x} \in \mathcal{D}: \mathbf{h}(\mathbf{x}) < \mathbf{0}} \Phi_\mu(\mathbf{x}) \qquad \text{s. t.} \qquad \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

- μ: barrier parameter.
- Fixed μ: solve an interior subproblem.
- **Decrease** μ: weaken the barrier away from the boundary.

Why use a logarithm?

On a convex domain, let $b_\mu(\mathbf{x}) = -\mu \log(-h(\mathbf{x}))$ for $h(\mathbf{x}) < 0$.

$$\nabla b_\mu = \frac{\mu}{-h}\nabla h, \qquad \nabla^2 b_\mu = \frac{\mu}{-h}\nabla^2 h + \frac{\mu}{h^2}\nabla h \nabla h^T.$$

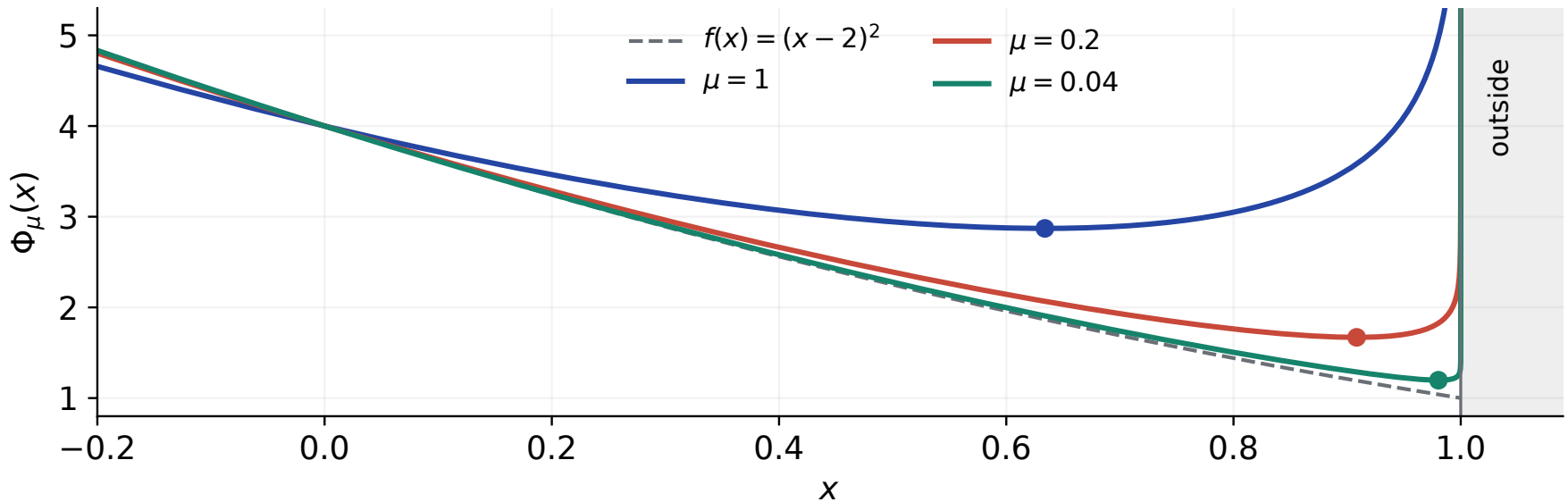
- Convex $h \Rightarrow$ convex b_μ ($\nabla^2 b_\mu \succeq 0$).
- Convex f, h_j's $\Rightarrow$ convex Φ_μ.
- Other walls, e.g., $\mu/[-h]$, are possible.

NB. Convex objective $\not\Rightarrow$ well-posedness.

A scalar example: attraction versus repulsion

$$\min_x (x-2)^2 \quad \text{s. t.} \quad x \le 1.$$

$$\Phi_\mu(x) = (x-2)^2 - \mu \log(1-x), \quad x < 1.$$



- Objective: attraction toward $x = 2$. Barrier: repulsion as $x \uparrow 1$.
- Smaller μ moves the minimum toward the boundary.

Solve the barrier subproblem

$$\Phi'_\mu(x) = 2(x-2) + \frac{\mu}{1-x} = 0, \qquad \Phi''_\mu(x) = 2 + \frac{\mu}{(1-x)^2} > 0.$$

Set $s = 1 - x > 0$. Then

$$-2(1+s) + \frac{\mu}{s} = 0 \quad \Longleftrightarrow \quad 2s^2 + 2s - \mu = 0.$$

The barrier domain requires the positive root:

$$\boxed{s(\mu) = \frac{\sqrt{1+2\mu} - 1}{2}, \qquad x(\mu) = \frac{3 - \sqrt{1+2\mu}}{2}.}$$

Strict convexity $\Rightarrow$ unique global barrier minimizer.

$x(\mu) < 1$ for every $\mu > 0$, and $x(\mu) \uparrow 1$ as $\mu \downarrow 0$.

Why must we lower μ?

At a fixed strict interior point $\mathbf{x} : \mathbf{h}(\mathbf{x}) < \mathbf{0}$,

$$\Phi_\mu(\mathbf{x}) - f(\mathbf{x}) = -\mu \sum_j \log(-h_j(\mathbf{x})) \longrightarrow 0 \qquad (\mu \downarrow 0).$$

- The barrier affects the *whole* strict interior.
- If $h_j < -1$, its term even lowers the objective.
- Decreasing μ removes interior distortion.

Why must we lower μ? (cont'd)

$$\Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu \sum_{j=1}^{p} \log(-h_j(\mathbf{x})), \qquad \mu > 0.$$

Differentiate one term:

$$\nabla[-\mu \log(-h_j)] = \frac{\mu}{-h_j} \nabla h_j.$$

Why must we lower μ? (cont'd)

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Differentiate one term:

$$\nabla[-\mu \log(-h_j)] = \frac{\mu}{-h_j} \nabla h_j.$$

Define

$$\text{multiplier estimate : } \nu_j(\mathbf{x}, \mu) = \frac{\mu}{-h_j(\mathbf{x})} > 0.$$

Then

$$\boxed{\nabla \Phi_\mu = \nabla f + \mathbf{C}\boldsymbol{\nu}.}$$

Why must we lower μ? (cont'd)

$$\min_{\mathbf{x}\in\mathcal{D}:\mathbf{h}(\mathbf{x})<\mathbf{0}} \Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu \sum_{j=1}^{p} \log(-h_j(\mathbf{x})), \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

Subproblem KKT gives

$$\nabla\Phi_\mu + \mathbf{A}\boldsymbol{\lambda} = \mathbf{0}, \qquad \mathbf{g} = \mathbf{0}.$$

Insert $\nabla\Phi_\mu = \nabla f + \mathbf{C}\boldsymbol{\nu}$ and $\nu_j = \mu/[-h_j(\mathbf{x})]$

Why must we lower μ? (cont'd)

$$\min_{\mathbf{x}\in\mathcal{D}:\mathbf{h}(\mathbf{x})<\mathbf{0}} \Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu\sum_{j=1}^{p}\log(-h_j(\mathbf{x})), \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

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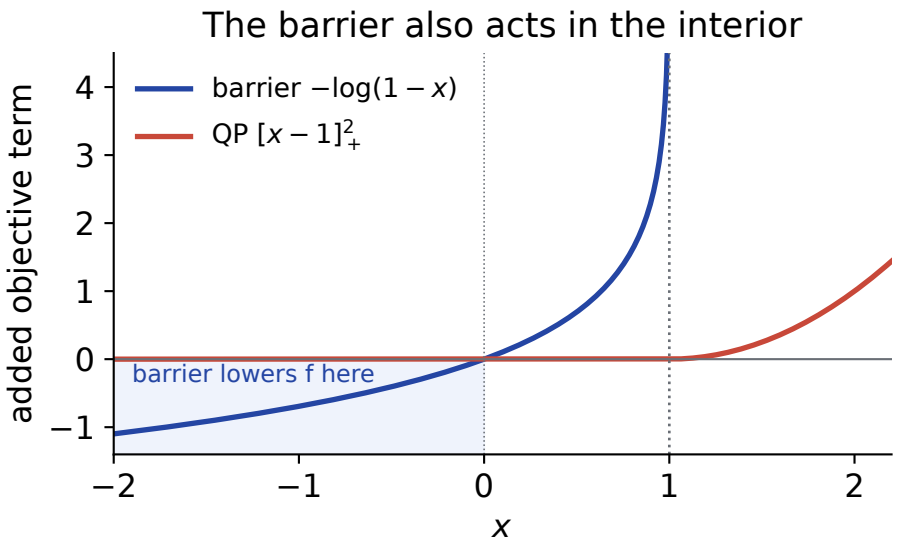
Insert $\nabla\Phi_\mu = \nabla f + \mathbf{C}\boldsymbol{\nu}$ and $\nu_j = \mu/[-h_j(\mathbf{x})] \Rightarrow$ **perturbed KKT**:

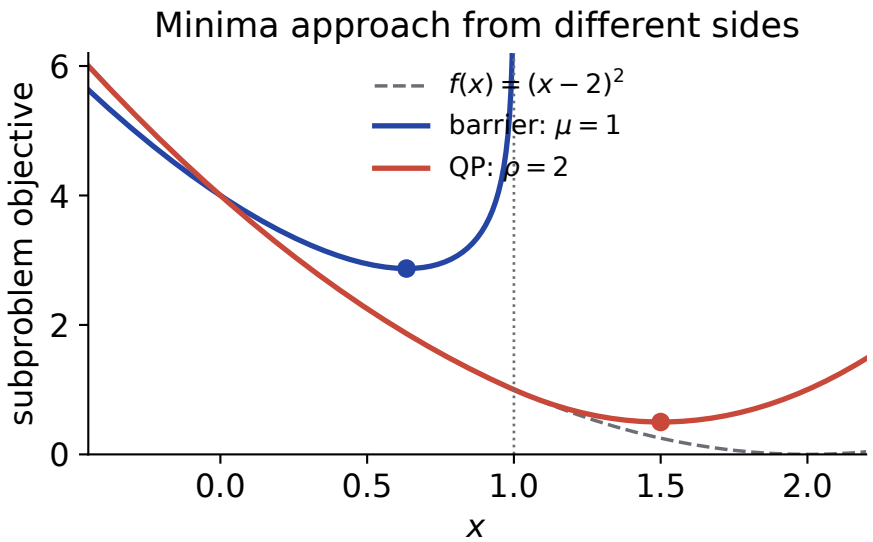
$$\boxed{\begin{array}{c} \nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu} = \mathbf{0}, \\ \mathbf{g} = \mathbf{0}, \qquad \mathbf{h} < \mathbf{0}, \qquad \boldsymbol{\nu} > \mathbf{0}, \\ -\nu_j h_j = \mu \quad (j = 1, \ldots, p). \end{array}}$$

- For $p \geq 1$, $\mu \to 0$ is necessary for exact KKT with these multipliers.

Comparing barrier and penalty

$$f(x) = (x-2)^2, \qquad h(x) = x - 1 \le 0.$$





Barrier: an interior wall. QP: an exterior violation cost.

They reshape different regions.

Barrier: $\mu \downarrow 0$. QP: $\rho \uparrow +\infty$.

A naive barrier method: pseudocode

(Assumes **the strict "interior" is not empty**.)
(Assumes each minimizer exists and can be found.)

Choose $\mu^{(0)} > 0$ and a fixed $0 < \theta < 1$.

1. At outer iteration k, solve exactly

$$\mathbf{x}^{(k+1)} \in \arg \min_{\mathbf{x} \in \mathcal{D}: \mathbf{h}(\mathbf{x}) < \mathbf{0},\, \mathbf{g}(\mathbf{x}) = \mathbf{0}} \left[f(\mathbf{x}) - \mu^{(k)} \sum_j \log(-h_j(\mathbf{x})) \right].$$

2. Set $\mu^{(k+1)} = \theta \mu^{(k)}$, and repeat.

Limitation 1: a strict starting point (phase I)

An auxiliary problem:

$$\min_{\mathbf{x}\in\mathcal{D},t} t \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) = 0, \qquad h_j(\mathbf{x}) \le t \quad (j = 1, \ldots, p).$$

$$\text{a feasible pair with } t < 0 \quad \Longrightarrow \quad \mathbf{x} \in \mathcal{S}^\circ.$$

- Stop at any feasible $t < 0$; no global solve is needed.
- If no equalities: SD/Armijo minimizes

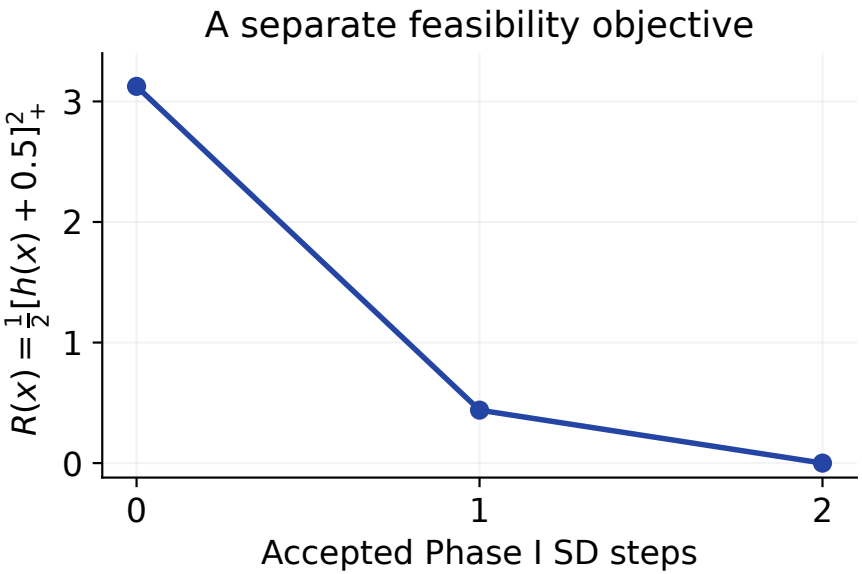
$$R_\eta(\mathbf{x}) = \tfrac{1}{2}\|[\mathbf{h}(\mathbf{x}) + \eta\mathbf{1}]_+\|_2^2, \quad \text{e.g., } \eta = \tfrac{1}{2},$$

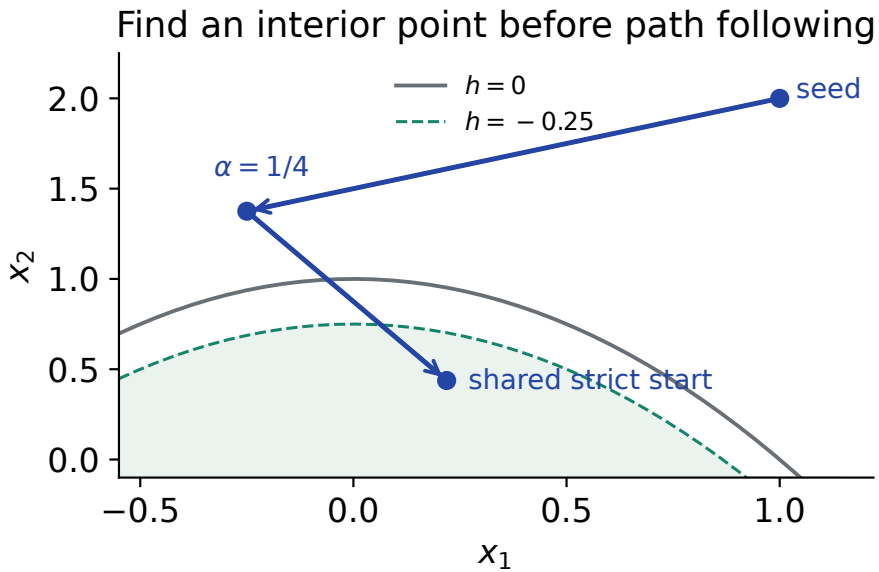
  stopping once every $h_j(\mathbf{x}) \le -\eta/2$.
- Search failure does not prove infeasibility.

Initialization for $h(\mathbf{x}) = x_1^2 + x_2 - 1$

From the infeasible seed $\mathbf{z}^{(0)} = (1, 2)^T$, SD/Armijo on $R = \frac{1}{2}[h + \frac{1}{2}]_+^2$ gives

$$(1, 2)^T \xrightarrow{\alpha=1/4} (-1/4, 11/8)^T \xrightarrow{\alpha=1} (7/32, 7/16)^T.$$





Final $h = -527/1024 < 0$.

Limitation 2: an inexact inner solver

Fix μ; warm-start from the previous outer point and solve to accuracy $\varepsilon_{\text{in}}^{(k)}$.

- **No equalities:** SD/Newton with line search for Φ_μ.

$$\mathbf{H}_\mu = \nabla^2 \Phi_\mu(\mathbf{x}), \qquad \mathbf{H}_\mu \mathbf{d} = -\nabla \Phi_\mu(\mathbf{x}).$$

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- **With equalities:** LN locally, or (damped) SQP with the ℓ_1 merit function

$$\begin{pmatrix} \nabla^2_{\mathbf{xx}} L_\mu & \mathbf{A} \\ \mathbf{A}^T & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{d} \\ \delta \end{pmatrix} = - \begin{pmatrix} \nabla \Phi_\mu + \mathbf{A}\boldsymbol{\lambda} \\ \mathbf{g} \end{pmatrix},$$

  where $L_\mu = \Phi_\mu + \boldsymbol{\lambda}^T \mathbf{g}$.

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At the returned strict point,

$$r_{\text{in}} = \max\{\|\mathbf{g}\|_\infty, \|\nabla \Phi_\mu + \mathbf{A}\boldsymbol{\lambda}\|_\infty\}.$$

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Then

$$r_{\text{stat}} = \|\nabla \Phi_\mu + \mathbf{A}\boldsymbol{\lambda}\|_\infty, \quad r_{\text{feas}} = \|\mathbf{g}\|_\infty, \quad r_{\text{dual}} = 0, \quad r_{\text{comp}} = \mu,$$

$$\Rightarrow \boxed{r_{\text{KKT}} = \max\{r_{\text{in}}, \mu\}.}$$

A full Newton step can cross the boundary

$$f(x) = (x-2)^2, \qquad h(x) = x - 1 \leq 0.$$

At $x = 0$, $\mu = 0.04$,

$$\Phi'_\mu(0) = -3.96, \qquad \Phi''_\mu(0) = 2.04, \qquad d = \frac{3.96}{2.04} \approx 1.941.$$

Full step: $x + d > 1$, so $\log(1 - x - d)$ is undefined.

One additional requirement: sufficient decrease + $\mathbf{h} < \mathbf{0}$.

Fraction-to-boundary for affine inequalities

If $h_j(\mathbf{x}) = \mathbf{a}_j^T\mathbf{x} - b_j < 0$, then

$$h_j(\mathbf{x} + \alpha\mathbf{d}) = h_j(\mathbf{x}) + \alpha\mathbf{a}_j^T\mathbf{d}.$$

If $\mathbf{a}_j^T\mathbf{d} > 0$, the boundary is reached at $-h_j/(\mathbf{a}_j^T\mathbf{d})$.

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If $\mathbf{a}_j^T\mathbf{d} > 0$, the boundary is reached at $-h_j/(\mathbf{a}_j^T\mathbf{d})$. Choose $0 < \tau < 1$:

$$\boxed{\alpha_{\max} = \min\left\{1,\ \tau \min_{\mathbf{a}_j^T\mathbf{d}>0} \frac{-h_j(\mathbf{x})}{\mathbf{a}_j^T\mathbf{d}}\right\}.}$$

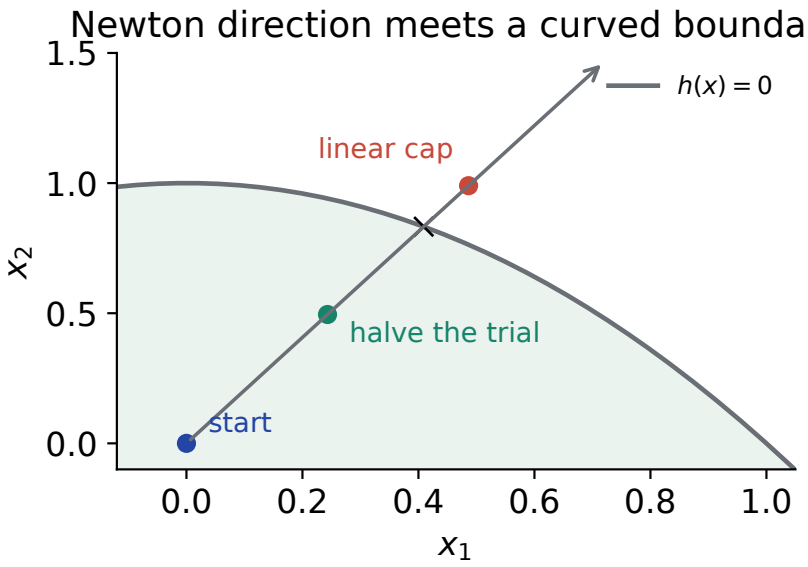
- If $\mathbf{a}_j^T\mathbf{d} > 0$ and $0 < \alpha \le \alpha_{\max}$, $h_j(\mathbf{x} + \alpha\mathbf{d}) \le (1 - \tau)h_j(\mathbf{x}) < 0$.
- If $\mathbf{a}_j^T\mathbf{d} \le 0$, this constraint imposes no upper bound. An empty inner minimum is $+\infty$.

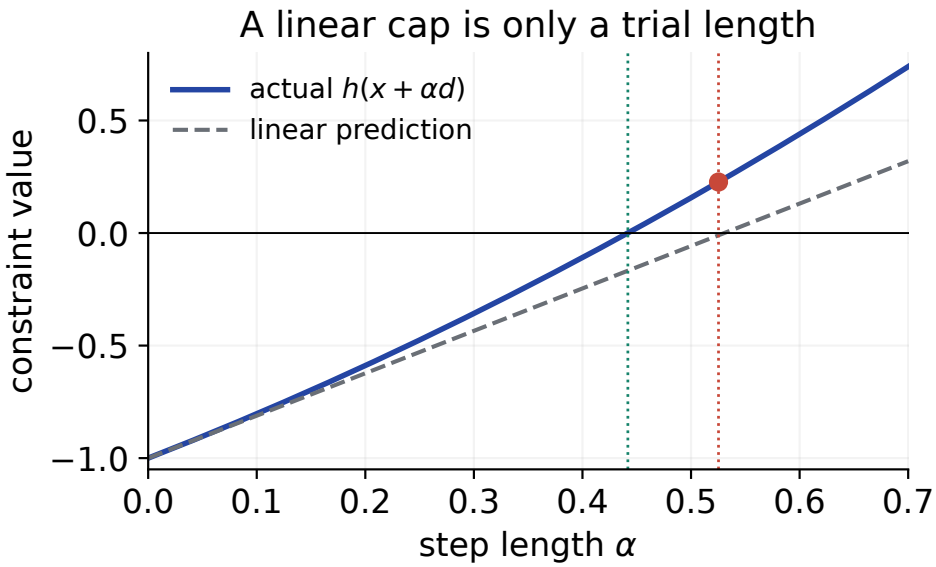
Curved boundaries need an actual feasibility check

$$f = \tfrac{1}{2}[(x_1 - 1)^2 + (x_2 - 2)^2], \quad h = x_1^2 + x_2 - 1 \le 0.$$

At $\mathbf{x} = (0, 0)^T$, $\mu = 0.04$, Newton gives $\mathbf{d} \approx (0.9259, 1.8846)^T$.

Linear prediction only: $h(\mathbf{x}) + \alpha \nabla h(\mathbf{x})^T \mathbf{d} < 0.$





Check actual $h_j(\mathbf{x} + \alpha \mathbf{d}) < 0$ before evaluating logarithms.

Limitation 3: (again!) ill-conditioning

Since $\nu_j = \mu/[-h_j(\mathbf{x})]$ and $\nabla\nu_j = (\mu/h_j^2)\nabla h_j$,

$$\nabla^2\Phi_\mu = \nabla^2 f + \sum_j \nu_j \nabla^2 h_j + \underbrace{\sum_j \frac{\mu}{h_j^2}\nabla h_j \nabla h_j^T}_{\text{positive semidefinite normal curvature}} .$$

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For an active constraint with $\nu_j(\mu) \to \nu_j^* > 0$,

$$-h_j(\mathbf{x}(\mu)) = \frac{\mu}{\nu_j(\mu)} = \Theta(\mu), \qquad \frac{\mu}{h_j(\mathbf{x}(\mu))^2} = \Theta(1/\mu).$$

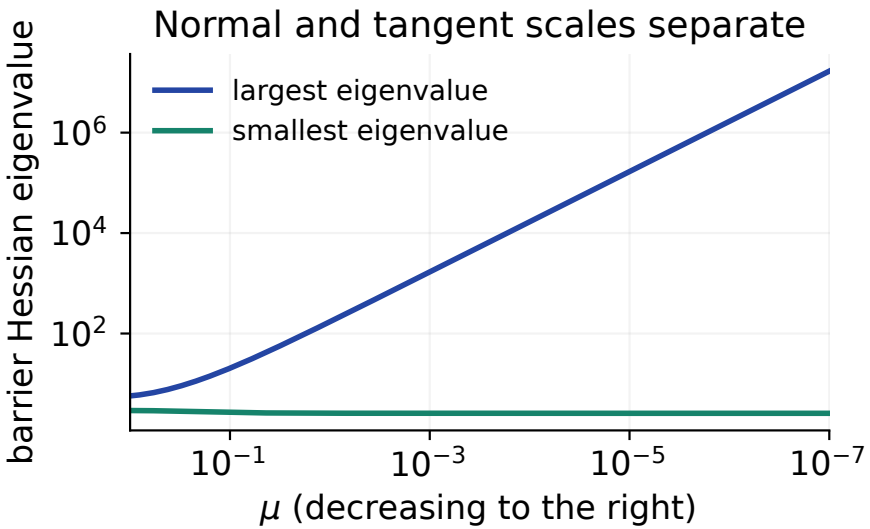
Large normal curvature can impair conditioning.

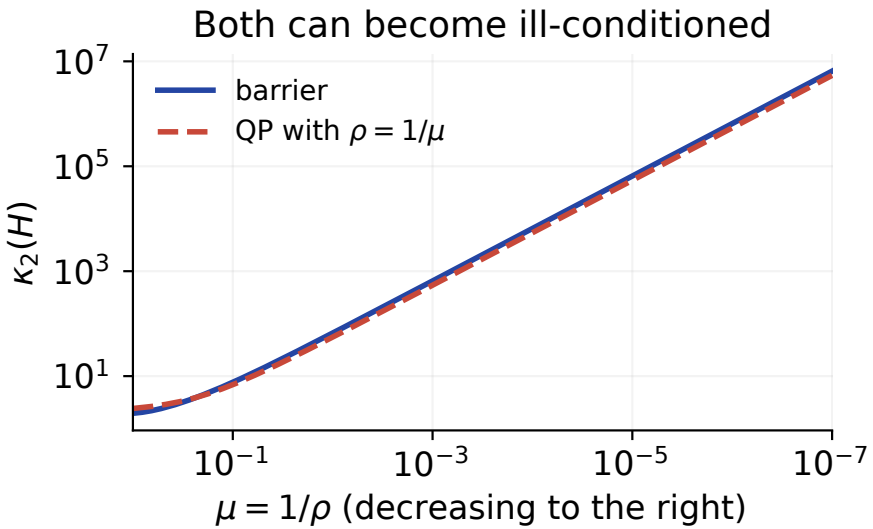
Normal versus tangent curvature

$$f = \tfrac{1}{2}[(x_1 - 1)^2 + (x_2 - 2)^2], \quad h = x_1^2 + x_2 - 1 \le 0.$$

At each method's exact subproblem solution, using its own ν and $\mathbf{a} = \nabla h = (2x_1, 1)^T$,

$$\mathbf{H}_{\text{barrier}} = \operatorname{diag}(1 + 2\nu, 1) + \frac{\nu^2}{\mu}\mathbf{a}\mathbf{a}^T, \qquad \mathbf{H}_{\text{QP}} = \operatorname{diag}(1 + 2\nu, 1) + \rho\mathbf{a}\mathbf{a}^T.$$





Practical barrier method

Choose a seed, $\mu^{(0)} > 0$, $0 < \theta < 1$, and fixed $0 < \varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} < \varepsilon_{\text{out}}$.

1. **Initialize:** Phase I finds $\mathbf{h}(\mathbf{x}^{(0)}) < 0$; choose equality multipliers if needed.
2. **Inner solve:** fix $\mu^{(k)}$, warm-start, and use domain-safe line search until
$$r_{\text{in}} \le \varepsilon_{\text{in}}^{(k)}.$$
3. **Recover multipliers:** set $\nu_j^{(k+1)} = \mu^{(k)} / [-h_j(\mathbf{x}^{(k+1)})]$.
4. **Check KKT:** stop if $r_{\text{KKT}} \le \varepsilon_{\text{out}}$. Otherwise set $\mu^{(k+1)} = \theta \mu^{(k)}$ and repeat.

Python: `barrier_method`

Common test problem

$$\min_{\mathbf{x}\in\mathbf{R}^2} \frac{1}{2}[(x_1-1)^2+(x_2-2)^2] \quad \text{s. t.} \quad h(\mathbf{x}) = x_1^2 + x_2 - 1 \le 0.$$

- Strict convexity + KKT $\Rightarrow$ unique solution:

$$\mathbf{x}^* \approx (0.312908409,\ 0.902088327)^T, \qquad \nu^* \approx 1.097911673.$$

- Unconstrained minimizer $(1,2)^T$: infeasible seed.

$$\text{Phase I} \quad \Longrightarrow \quad \text{shared start } (7/32, 7/16)^T.$$

Barrier / QP / ALM: same KKT test and Newton/Armijo inner solver.

Common test: fixed settings

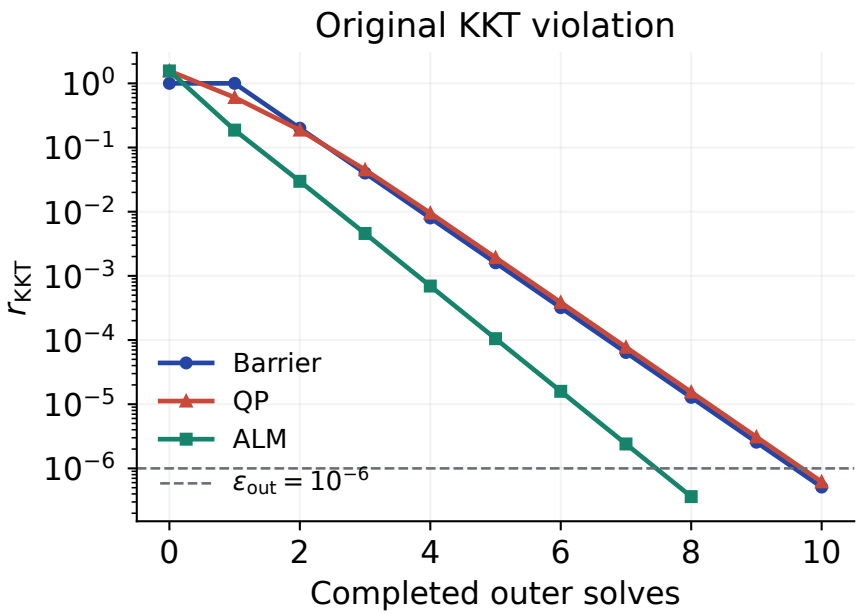
Shared start $(7/32, 7/16)^T$; warm-started Newton/Armijo.

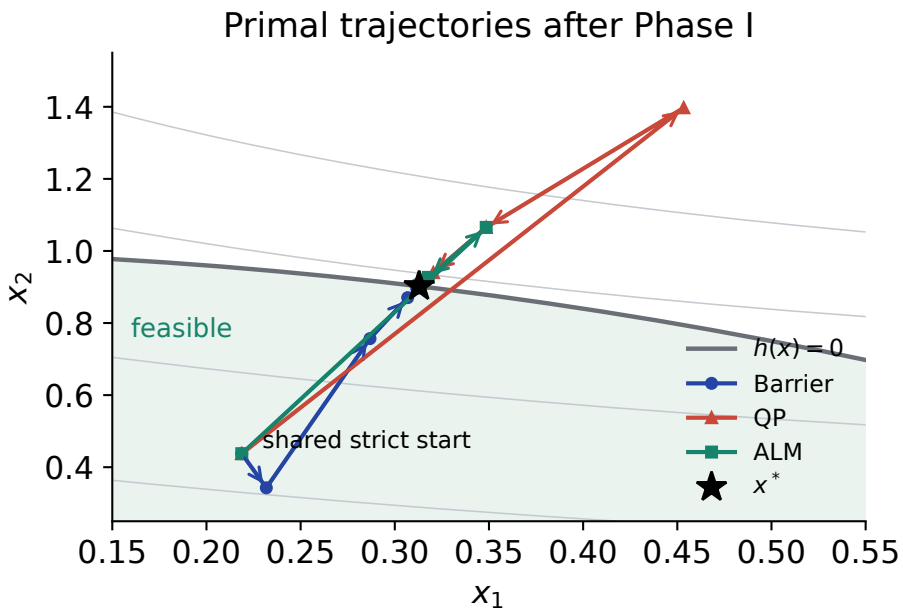
$$\varepsilon_{\text{in}}^{(k)} \equiv 10^{-7}, \qquad \varepsilon_{\text{out}} = 10^{-6}, \qquad \sigma = 10^{-4}, \quad \beta = 1/2.$$

Method	Initial parameter	Outer update
Barrier	$\mu^{(0)} = 1$	$\mu^{(k+1)} = 0.2\mu^{(k)}$
QP	$\rho^{(0)} = 1$	$\rho^{(k+1)} = 5\rho^{(k)}$
ALM	$\nu^{(0)} = 0,\ \rho = 5$	Multiplier update; fixed ρ

- Barrier: linear fraction $\tau = 0.99$ + actual domain checks.

Common test: KKT violation and primal trajectories





Shared strict start; plotted points follow successful inner solves.
Here barrier stays inside; QP / ALM approach the boundary from outside.

Common test: measured work and final residuals

	Barrier	QP	ALM
Outer solves / Newton steps	10 / 82	10 / 33	8 / 19
Stationarity r_{stat}	9.511×10^{-12}	2.614×10^{-10}	6.340×10^{-12}
Primal feasibility r_{feas}	0	5.621×10^{-7}	3.307×10^{-7}
Dual feasibility r_{dual}	0	0	0
Complementarity r_{comp}	5.120×10^{-7}	6.172×10^{-7}	3.631×10^{-7}
Overall r_{KKT}	5.120×10^{-7}	6.172×10^{-7}	3.631×10^{-7}

- Phase I for initialization: 2 SD steps, excluded above.
- Final parameters: barrier $\mu = 5.12 \times 10^{-7}$, QP $\rho = 1\,953\,125$, ALM $\rho = 5$.

Different priorities

Barrier: maintain strict inequalities in the main iterations.
QP / ALM: allow violation while approaching KKT.

Extension to primal–dual IPMs

IPM: interior-point method.

- **(Primal) barrier method:** solve in $\mathbf{x}$ and equality multipliers; substitute $\nu_j = \mu/[-h_j(\mathbf{x})]$.
- **Primal IPM:** work mainly with $\mathbf{x}$ while keeping $\mathbf{h}(\mathbf{x}) < \mathbf{0}$; barrier methods are one instance.
- **Primal–dual IPM:** treat $\boldsymbol{s}, \boldsymbol{\nu}$ as separate variables; apply Newton to

$$\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu} = \mathbf{0}, \quad \mathbf{g} = \mathbf{0},$$

$$\mathbf{h} + \boldsymbol{s} = \mathbf{0}, \quad s_j \nu_j = \mu, \quad \boldsymbol{s}, \boldsymbol{\nu} > \mathbf{0}.$$

SQP and IPMs: two widely used approaches for constrained optimization.

Take-home messages

Methods comparison: inner problems

Method	Inner problem	Constraint information
QP	Unconstr. penalty model ($\rho^{(k)}$)	$\boldsymbol{\lambda}^{(k+1)} = \rho^{(k)}\mathbf{g}(\mathbf{x}^{(k)})$
ALM	Unconstr. augmented Lagrangian (ρ)	$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho\mathbf{g}(\mathbf{x}^{(k)})$
SQP	Quadratic program (quadratic obj. + linear constrs.)	$\mathbf{H}_L^{(k)}$ (obj.) + $\widehat{\boldsymbol{\lambda}}^{(k+1)}$ (constrs.)
Barrier	Log barrier w. equality constrs. ($\mu^{(k)}$)	$\mathbf{h}(\mathbf{x}^{(k)}) < 0$, $\nu_j^{(k)} = \frac{\mu^{(k)}}{-h_j(\mathbf{x}^{(k)})}$

Take-home messages (cont'd)

Methods comparison: strengths and main practical issues

Method	Useful feature	Main practical issue
QP	Simple unconstr. subprob.	$\rho^{(k)} \uparrow +\infty$

Take-home messages (cont'd)

Methods comparison: strengths and main practical issues

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Take-home messages (cont'd)

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SQP	Model and step safeguards Local fast convergence	Well-posedness; merit; complementarity

Take-home messages (cont'd)

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QP	Simple unconstr. subprob.	$\rho^{(k)} \uparrow +\infty$
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SQP	Model and step safeguards Local fast convergence	Well-posedness; merit; complementarity
Barrier	Strict inequalities No active-set choice	Initialization; boundary steps; $\mu^{(k)} \downarrow 0$

Common for all: balance between inner and outer levels.

Passing to a KKT limit

Assume $\mu^{(k)} \to 0$, vanishing inner errors, and

$$(\mathbf{x}^{(k)}, \boldsymbol{\lambda}^{(k)}, \boldsymbol{\nu}^{(k)}) \to (\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*), \qquad \mathbf{x}^* \in \mathcal{D}.$$

Continuity and $-\nu_j^{(k)} h_j(\mathbf{x}^{(k)}) = \mu^{(k)}$ imply, at $\mathbf{x}^*$,

$$\nabla f + \mathbf{A}\boldsymbol{\lambda}^* + \mathbf{C}\boldsymbol{\nu}^* = 0, \qquad \mathbf{g} = 0,$$
$$\mathbf{h} \le 0, \quad \boldsymbol{\nu}^* \ge 0, \quad \nu_j^* h_j = 0.$$

- Primal convergence + LICQ at the limit $\Rightarrow$ multiplier convergence.
- Near a KKT solution: LICQ + strict complementarity + SOSC $\Rightarrow$ a local central path; no global uniqueness claim.
- Nonconvex problem: KKT need not mean a minimum.

Complementarity: component, sum, and norm

For $\mathbf{h} < 0$ and $\nu_j = \mu/(-h_j)$,

$$-\nu_j h_j = \mu, \qquad -\boldsymbol{\nu}^T \mathbf{h} = p\mu, \qquad \|\boldsymbol{\nu} \odot \mathbf{h}\|_\infty = \mu.$$

- No inner stationarity needed for these identities.
- They measure complementarity; stationarity needs its own check.

On a convex domain, with convex f, h_j, affine equalities, and an *exact* barrier KKT point,

$$\boxed{0 \leq f(\mathbf{x}) - f^* \leq p\mu.}$$

The objective-error bound needs both convexity and exact stationarity.